

MOHAMMAD SAMEER MALEK

Forward-Deployed AI Engineer | Co-Founder & CIO, Orchestrator AI | Applied AI, Computer Vision, AI Solutions
Calgary, AB | maleksameer715@gmail.com | +1 (825) 288-2786 | [LinkedIn](#) | [Portfolio](#) | [GitHub](#)

PROFESSIONAL PROFILE

- Forward Deployed Engineer (FDE) by practice: I embed inside client businesses, take ambiguous problems through discovery, architecture, build, and deployment, and stay until the system runs the business, because most AI projects die in the gap between the demo and production.
- Applied AI on two fronts: agentic LLM systems (multi-agent research pipelines, MCP tool design, RAG, eval-gated synthesis) and computer vision (YOLOv11, RF-DETR, DINOv3 training, OCR and document intelligence), matching the pattern to the problem instead of defaulting to an agent for everything.
- Operate 58 production automation workflows executing more than 78,000 times per month across accounts payable, CRM, lead capture, and reporting, built because the alternative was hiring people to do data entry.
- Security and trust as defaults: SOC 2 Type II program ownership, RBAC, PII/PHI redaction, and per-tenant encrypted credentials, because enterprise clients buy trust before they buy features.

SKILLS

- **AI / LLM Agents:** Claude API (Anthropic), OpenAI API, Google Gemini and Vertex AI, agentic patterns (orchestrator-workers, parallelization, evaluator-optimizer), MCP (Model Context Protocol) tool design, RAG, prompt engineering, agent evaluations (LLM-as-judge, regression and capability suites, RAGAS-style gates), LangChain, structured outputs, human-in-the-loop design.
- **Computer Vision / ML:** YOLOv11, RF-DETR, DINOv3, object detection, instance segmentation, model training and fine-tuning, data annotation (Roboflow), OCR (PaddleOCR, Tesseract, LlamaParse), OpenCV, scikit-learn, speech-to-text (Whisper, AssemblyAI), vector stores (FAISS, Pinecone, Supabase).
- **Languages & Frameworks:** Python, TypeScript, JavaScript, React, Next.js, Node.js, Express, FastAPI, tRPC, Prisma, Drizzle, SQLAlchemy, Zod.
- **Cloud, Infra & Data:** GCP (Cloud Run, Cloud SQL, Cloud Functions, Secret Manager, Artifact Registry), Docker, Kubernetes (GKE), Nginx, Vercel, Firebase, Supabase, Microsoft Azure, GitHub Actions CI/CD, Datadog (RUM, logs), PostgreSQL, MongoDB, Redis.
- **Integrations & Practice:** QuickBooks Online, Maximizer CRM, Microsoft Graph and M365 SSO, Google Workspace, Stripe, n8n, OAuth2, webhook architecture; multi-tenant SaaS, RBAC design, SOC 2 Type II, OWASP audits, event-driven architecture, token and cost optimization.

PROFESSIONAL EXPERIENCE

Co-Founder & CIO, Forward-Deployed AI Engineering | Orchestrator AI Systems Ltd., Calgary June 2025 - Present

Calgary AI consultancy built on a forward-deployed model: embed with the client, own the problem from discovery through deployment, ship production systems.

- Embedded with William Blake Homes (construction) to turn a loosely specified estimating idea into Keystone, a production estimating platform (React, Node, Prisma, PostgreSQL, Microsoft 365 SSO). Discovery corrected the client's own PRD (11 cost divisions, not 12; DeskReview, not Bluebeam) and produced a 68-item prioritized v2 backlog, because scope errors are far cheaper to catch before build than after.
- Delivered an enterprise RAG deployment for ZS2 with VPC isolation, RBAC, PII/PHI redaction, and full audit logging, because the data could not leave the client perimeter and compliance needed to see every retrieval.
- Directed EstimAgent, an AI construction cost estimator: trained YOLOv11, RF-DETR, and DINOv3 detection and segmentation models on over 4,200 annotated floorplans (Roboflow) with a multi-stage extraction pipeline (vector PDF geometry, PaddleOCR and Tesseract, flood-fill room segmentation, scikit-learn page classification), because takeoffs were the slowest and most error-prone step in the client's estimating.
- Built AutoPilot, a browser automation platform on Claude Vision and Playwright: human-in-the-loop escalation whenever confidence drops below 40%, batch runs exceeding 500 rows with auth-state reuse, and per-tenant encrypted credentials with audit trails on Cloud Run, because unattended agents need explicit trust boundaries and a human path when they are unsure.
- Designed a production MCP server exposing more than 45 Maximizer CRM tools (TypeScript, OAuth2, stdio and HTTP transports), consolidating multi-step CRM operations into a small set of high-signal, strictly schematized tools rather than raw endpoint wrappers, because agents perform best with purposeful tools that conserve context.
- Own the SOC 2 Type II readiness program (policy authorship, Google Workspace and Chrome Enterprise controls, vendor DPA risk) and price AI features with per-token cost models in client proposals, because enterprise buyers ask about trust and predictable cost before the first demo.
- Shipped computer vision and 3D delivery for an energy-sector client (drone surveillance console on Cloud Run) and self-hosted image-to-3D generation on GCP GPU VMs benchmarked across L4 and A100 instances, because compute cost is a design constraint, not an afterthought.

Embedded AI & Automation Lead (Forward-Deployed) | Sonata Design, Calgary May 2025 - Present

Long-running embedded engagement: in practice the forward-deployed engineer for a multi-brand design and manufacturing group, owning its AI, automation, and product stack.

- Built FabricLab (fabriclab.ai), a multi-tenant SaaS quoting and work-order platform (4 calculation engines, BOM generation, 4-tier RBAC, Stripe billing, Gemini fabric visualization, Datadog RUM; over 34 users across 4 organizations), then forward-deployed a dedicated instance for Sheer Illusions with hands-on onboarding, workroom-specific pricing, and on-site iteration until their estimators quoted through it daily, because a platform only counts once the client runs their business on it.
- Shipped a 7-phase invoice automation pipeline processing 700 to 1,000 invoices per month (Gmail intake, LlamaParse OCR, schema-driven LLM extraction, Supabase fuzzy vendor matching, QuickBooks Online posting) with SHA256 de-duplication and vendor-specific handling, including Hunter Douglas multi-invoice PDFs (5 to 20 per document), scalable past 100 vendors through schema-driven onboarding, because vendor formats drift and a pipeline that cannot adapt becomes shelfware within a quarter.
- Operate 58 production n8n workflows executing over 78,000 times per month with a success rate above 98% across AP, CRM sync, lead capture, and reporting, eliminating more than 320 hours of manual work monthly; the largest, a 179-node purchase-order pipeline, automated roughly 90% of one department and took it from 2 full-time staff to 1, because the alternative was hiring for data entry.
- Treat workflows like evaluated agents: centralized failure categorization with execution-linked logs and alerts, parser changes validated against more than 40 documented real-message test cases before shipping, and regressions root-caused from execution transcripts, including a runaway loop that silently kept reprocessing a single purchase order until the workflow wiring fault was isolated, because guess-and-check debugging does not scale.
- Designed Meridian, a multi-provider deep-research engine (Python, FastAPI): parallel provider fan-out into an evidence-linked claim graph, synthesis grounded strictly in cited claims, and a RAGAS-style evaluation gate before any deliverable ships; it powers BIOP, a broker-intelligence web application whose model council runs 5 parallel research agents plus a synthesizer, because business decisions cannot rest on hallucinated citations and single-model research reads shallow.
- Built ProQuote, a quote-to-e-signature web application (React, Node, Express, PostgreSQL, Drizzle) that normalizes supplier quotes from 7 vendor formats through a single parser registry, layered with a Gemini and Vertex AI measure-sheet OCR extractor, Maximizer CRM and CompanyCam integrations, and an audit-trailed approval-token signature flow, because sales retyped supplier quotes line by line and per-vendor parsers broke every time a template changed.
- Shipped Mozart, an org-wide RAG assistant over Google Drive (FAISS, Firestore, Cloud Functions), alongside lead automations for Brookfield, Virtuo, and Somfy with LLM-generated follow-up emails, because institutional knowledge sat buried in shared folders and inbound leads waited overnight for a first reply.
- Built Cadence (heycadence.ai), a meeting transcription web application running self-hosted Whisper inside a private VPC with diarized Gemini summaries, because meeting audio is sensitive and should never transit a third-party service.

EDUCATION

Southern Alberta Institute of Technology (SAIT), Calgary, AB May 2023 - December 2024

Technical Diploma in Software Development, CGPA 3.72 / 4.00

CERTIFICATIONS

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- Microsoft Azure Fundamentals (AZ-900), Microsoft, January 2025.
 - Introduction to Google Cloud Platform, SimpliLearn, May 2025.
 - n8n Course: No Code AI Agent Builder, SimpliLearn, May 2025.